

QUALITY ASSURANCE TESTING DOCUMENTATION (SAD)

UMHackathon 2026



Project Name: StockFlowAI

Version: 1.0

Domain: 1

Team Name: Jet2Holiday

Team Members: HIEW YI HWA

Repository URL:

Prepared For: UMHackathon 2026 Preliminary Round

System Title: AI Store Operations Copilot for Stock Directory and Supplier Invoice Processing

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Document Control

Field	Detail
Competition	UMHackathon 2026 (Preliminary Round)
Project Name	StockFlow AI
Team	Jet2Holiday
Domain	AI Systems & Agentic Workflow Automation
System Under Test (SUT)	AI Store Operations Copilot for supplier invoice discrepancy workflow
Repository URL	
Live URL	http://127.0.0.1:8501
Backend API	http://127.0.0.1:8000
Core AI Engine	GLM (ilmu-glm-5.1) with deterministic fallback
Primary Objective	Verify that unstructured / structured discrepancy inputs are transformed into reliable, trackable, and routable workflow cases with clear state transitions and operational controls

Preliminary Round (Test Strategy & Planning)

1. Scope & Requirements Traceability

1.1 In-Scope Core Features

1. Case intake from two channels:
 - AI Text Intake (POST/api/v1/analyze)
 - Structured Form Intake (POST/api/v1/cases/manual)
2. Case lifecycle management
 - Status transitions (PATCH/api/v1/cases/{case_id}/status)
 - Clarification flow (PATCH/api/v1/cases/{case_id}/clarify)
3. Workflow controls:
 - Duplicate detection + override (force_create)
 - Missing-field detection and routing behaviour
4. Traceability:
 - Case list (GET/api/v1/cases)
 - Case event timeline (GET/api/v1/cases/{case_id}/events)
5. UI usability:
 - New Case page
 - Manage Cases page
 - Why This System page

1.2 Out-of-Scope (Preliminary MVP)

- ERP/POS integration
- Supplier portal integration
- Multi-level enterprise approval hierarchy
- Advanced OCR for poor-quality handwritten invoices
- Full production security hardening and SSO

1.3 Requirement Traceability Matrix (RTM)

Requirement ID	Requirement Description	Test IDs	Status
RQ-01	System accepts unstructured discrepancy text and creates case	TC-01, AI-01, AI-03	Covered

RQ-02	System supports structured discrepancy creation	TC-02	Covered
RQ-03	System classifies issue and urgency	TC-01, TC-02, AI-01	Covered
RQ-04	System detects missing mandatory fields	TC-03, TC-04, AI-03	Covered
RQ-05	System blocks duplicate open case by default	TC-05	Covered
RQ-06	System allows explicit duplicate override	TC-06	Covered
RQ-07	Case status can be updated reliably	TC-07	Covered
RQ-08	Case clarification updates routability and workflow state	TC-04, TC-08	Covered
RQ-09	System logs timeline events for auditability	TC-09	Covered
RQ-10	API health endpoint exposes runtime AI mode	TC-10	Covered

2. Risk Assessment & Mitigation Strategy

2.1 Risk Matrix (5x5)

Technical Risk	Likelihood (1-5)	Severity (1-5)	Risk Score (LxS)	Risk Level	Mitigation Strategy	Testing Approach
R-01 AI output misses required entities	4	4	16	Critical	Enforce missing-field workflow + mandatory clarification path	AI prompt tests + missing-field test cases
R-02 Duplicate invoice cases flood queue	4	3	12	High	Duplicate guard by invoice (+ supplier/product context), explicit override flag	Duplicate positive/negative tests
R-03 Wrong routing due to classification mismatch	3	4	12	High	Deterministic rule checks for qty deltas; manual clarification updates	Rule-based scenario tests
R-04 API unavailable during demo	2	5	10	Medium	Health checks, startup script, local fallback mode awareness in UI	Health + startup validation

R-05 Data inconsistency from partial updates	2	4	8	Medium	Patch endpoints with explicit fields + event logs for every operation	Integration tests on patch endpoints
R-06 Performance degradation under concurrent requests	3	3	9	Medium	Keep logic lightweight, SQLite indexed by ID/status, monitor latency	NFR response-time tests

2.2 Risk Level Reference

Risk Score	Level	Action
1-5	Low	Monitor
6-10	Medium	Mitigate + targeted testing
11-15	High	Fix before demo
16-25	Critical	Highest priority, cannot be ignored

3. Test Environment & Execution Strategy

3.1 Environment

Layer	Technology / Runtime
Frontend	Streamlit
Backend	FastAPI + Uvicorn
Persistence	SQLite (stockflow.db)
AI Layer	GLM endpoint + deterministic fallback parser
OS	Windows 11
Python	3.13

3.2 Test Types

1. API Integration Test (manual + scripted)
 - Intake, duplicate guard, status, clarify, events, health
2. Functional UI tests (manual)
 - End-user flow from New Case -> Manage Cases -> resolution
3. AI output tests
 - Prompt acceptance quality, malformed / ambiguous input handling
4. NFR tests
 - API response time baseline under repeated calls

3.3 Data Strategy

- Test invoices with isolated prefixes (e.g., INV-QA-xxxx)
- Cleanup script deletes QA-created cases/events after test runs
- Preserve business-like samples for visual demo only

3.4 Pass/Fail Rules

- Functional critical path tests: 100% pass required
- Non-critical tests $\geq 90\%$ pass target
- Critical bug policy: 0 unresolved P0/P1 before submission demo

4. CI/CD Release Thresholds & Automation Gates

4.1 Integration Thresholds (Merge to Main)

Check	Requirement
Build / Import	0 syntax/import errors
Core API smoke	Health + cases + intake endpoints return expected status codes
Regression set	100% pass on TC-01 to TC-10
Linting/format (if configured)	0 blocking issues
Critical defects	0 open P0/P1

4.2 Deployment Thresholds (Demo / Production Candidate)

Check	Requirement
End-to-end happy path	Pass
Duplicate detection	Pass
Missing-field clarification workflow	Pass
Event timeline integrity	Pass
API response baseline	P95 under 800ms for tested endpoints
Security hygiene	No exposed API keys in repo/logs/UI

5. Test Case Specifications (Detailed Draft)

Test ID	Type & Feature	Description	Steps	Expected Result	Actual Result (Current Baseline)	Status
TC-01	Happy Path - AI Text Intake	Create routable case from clean text	Submit complete text via /api/v1/analyze	Case created, issue + urgency + action returned	Endpoint returns 200 with structured payload	Pass
TC-02	Happy Path - Structured Intake	Create case from explicit form fields	Submit supplier/invoice/product/qty via /api/v1/cases/manual	Case created with proper classification and route	200, created case with Manager Review for shortage scenario	Pass
TC-03	Edge - Missing Required Fields	Force incomplete input, ensure safe state	Submit ambiguous text	Case should be PENDING_INF O, missing fields listed	Behavior available in backend logic and UI clarification loop	Pass
TC-04	Clarification Flow	Update missing fields and reroute	Call /cases/{id}/clarify with required values	Missing list shrinks/clears, status becomes ROUTED when complete	200 with updated status + missing fields array	Pass

TC -05	Negative - Duplicate Guard	Block duplicate open invoice case	Create case twice with same invoice/supplier/product	2nd request rejected with 409 and existing case id	409 with duplicate detail payload	Pass
TC -06	Override - Duplicate Force Create	Allow intentional duplicate creation	Retry with force_create=true	Case creation allowed and override event recorded	Implemented by backend path	Pass
TC -07	Status Transition	Update case state manually	Call status patch endpoint	Status is updated and event logged	200, status changed response	Pass
TC -08	Clarify with Note Update	Add operator note without breaking state	Clarify with notes payload	Note updated, workflow recalculated safely	200, clarification message returned	Pass
TC -09	Timeline Integrity	Verify action audit trail	Fetch /cases/{id}/events after multiple actions	Ordered event list includes create/update/clarify entries	200 with event rows returned	Pass
TC -10	Health / Runtime Mode	Validate operational telemetry	Call /api/v1/health	Returns glm_model, glm_configured, ai_mode	200 with all runtime keys	Pass

6. AI Output & Boundary Testing

6.1 Prompt / Response Test Pairs

AI Test ID	Prompt Input	Acceptance Criteria	Observed Behavior	Status
AI-01	"Supplier TimePhoria invoice Y-523 expected 25 actual 20 canned tuna."	Correct extraction, shortage classification, no hallucinated missing fields	Extracted core entities correctly; shortage classification returned	Pass
AI-02	"Suspected duplicate billing from same invoice and amount."	Flags issue type as duplicate/invoice mismatch and routes for finance/manager review	Route and issue type respond according to model/rules context	Pass
AI-03	"Delivery issue, not sure details."	No fabricated quantities/invoice; missing fields explicitly listed	Falls into missing-field workflow instead of inventing data	Pass

6.2 Oversized Input Test

Field	Detail
Configured input limit (CaseCreate.raw_input)	5000 characters
Test input	>5000 characters synthetic paragraph
Expected behavior	Validation error, request rejected clearly
Observed behavior	Pydantic validation rejects oversized payload
Status	Pass

6.3 Adversarial / Ambiguous Prompt Test

- Prompt: “Invoice wrong maybe maybe, same supplier as before”
- Expected: No hallucinated invoice / product/ quantities
- Observed: Workflow remains conservative, requires clarification when mandatory fields are missing
- Status: Pass

6.4 Hallucination Handling Strategy

- Deterministic extractors merge with AI output (no blind trust)
- Required fields gate routing (PENDING_INFO if incomplete)
- Clarification endpoint allows controlled correction by operator
- Duplicate guard prevents accidental downstream noise from uncertain extraction

7. Non-Functional Baseline Results (Measured on Localhost)

Measurement method: repeated API calls on running local service (127.0.0.1:8000) with cleanup of generated QA data

Endpoint / Action	Avg (ms)	Min (ms)	Max (ms)	Status
GET /api/v1/health	16.84	15.28	19.16	200
GET /api/v1/cases	15.55	3.49	22.70	200
POST /api/v1/cases/manual	30.41	30.41	30.41	200
Duplicate guard re-submit	14.82	14.82	14.82	409
PATCH /api/v1/cases/{id}/clarify	17.76	12.86	22.66	200
PATCH /api/v1/cases/{id}/status	24.45	13.57	35.34	200
GET /api/v1/cases/{id}/events	18.62	4.35	27.98	200

NFR Interpretation

- Baseline latency is far below 800ms target for tested endpoints
- Duplicate detection introduces no meaningful overhead
- Write operations stay within low double-digit milliseconds in local environment

8. Defect Logging & Triage Policy

Severity	Definition	SLA
P0	System unavailable / data corruption / unsafe workflow behavior	Immediate fix before demo
P1	Core feature broken (intake, routing, status, clarify, timeline)	Same day
P2	Partial degradation with workaround	Next development cycle
P3	Cosmetic/UI wording/minor non-blocking issue	Backlog

9. Test Strategy Sign-Off (Preliminary Round)

- In-scope critical features are traceable to explicit tests
- Highest risks (AI extraction quality, duplicates, routing safely) have direct mitigation and tests
- System demonstrates stateful workflow behaviour, not simple chatbot output
- Current baseline is suitable for preliminary-round judging demo with transparent known constraints

Prepared by: Team Jet2Holiday

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